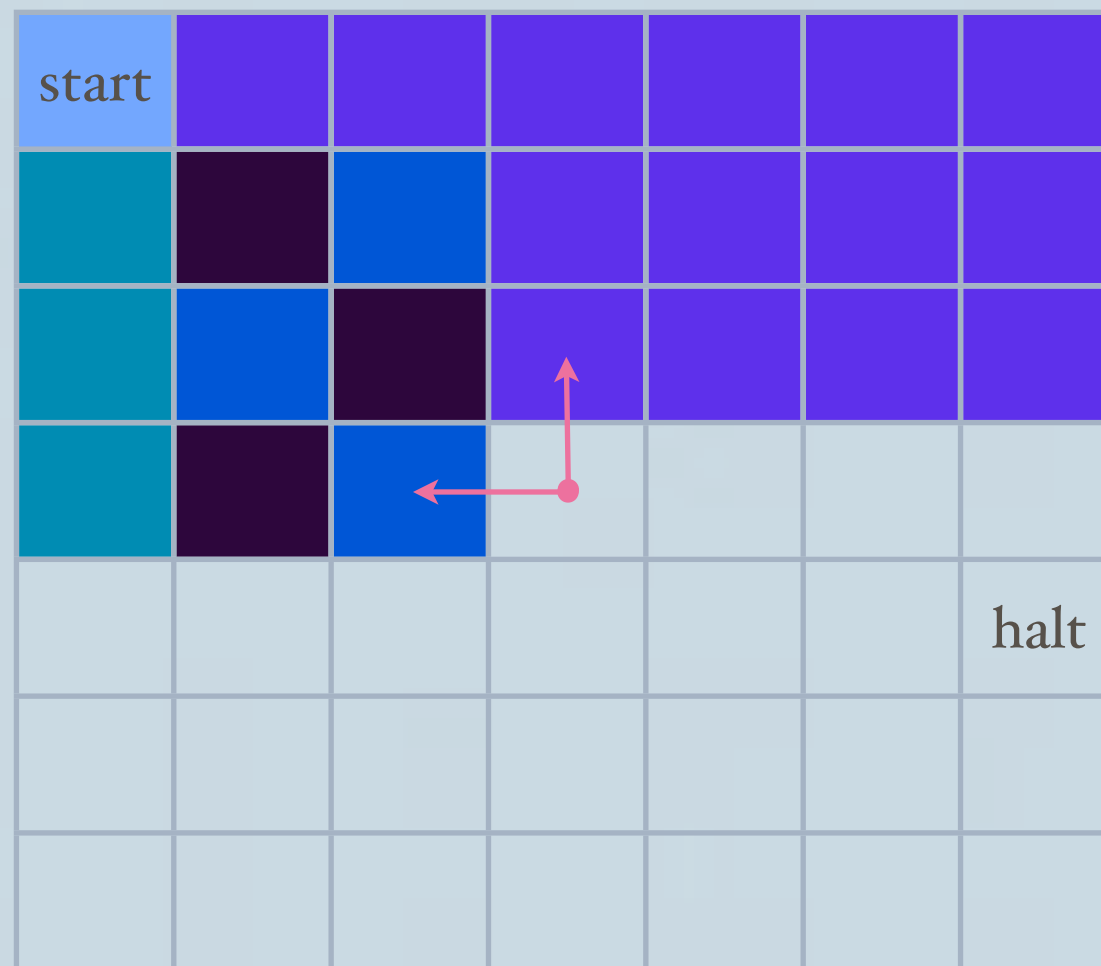


Backup Slides

Finite Undecidability

Theorem: It is undecidable to check given $I \in \forall^*\exists^*$ and $\delta \in \text{EPR}$ whether I is inductive for δ over **finite structures**.



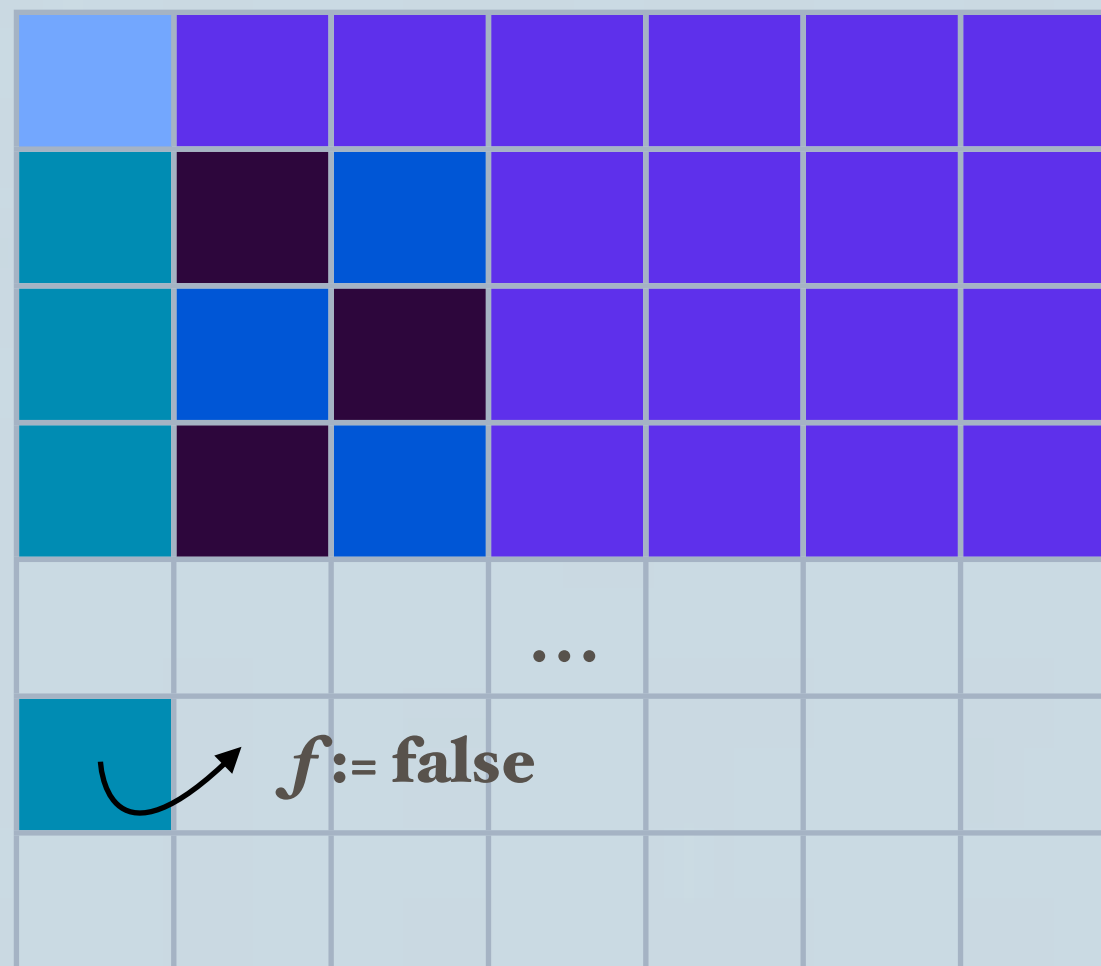
Invariant: Valid tiling, and “halt” cannot be placed

General Undecidability

Theorem: It is undecidable to check given $I \in \forall^* \exists^*$ and $\delta \in \text{EPR}$ whether I is inductive for δ over **general** (finite or infinite) **structures**.

$H(0) = \text{true}$
 $H(1) = \text{true}$
 $H(2) = \text{true}$
 $H(3) = \text{true}$
 $\dots$

$H(\omega) = \text{false}$

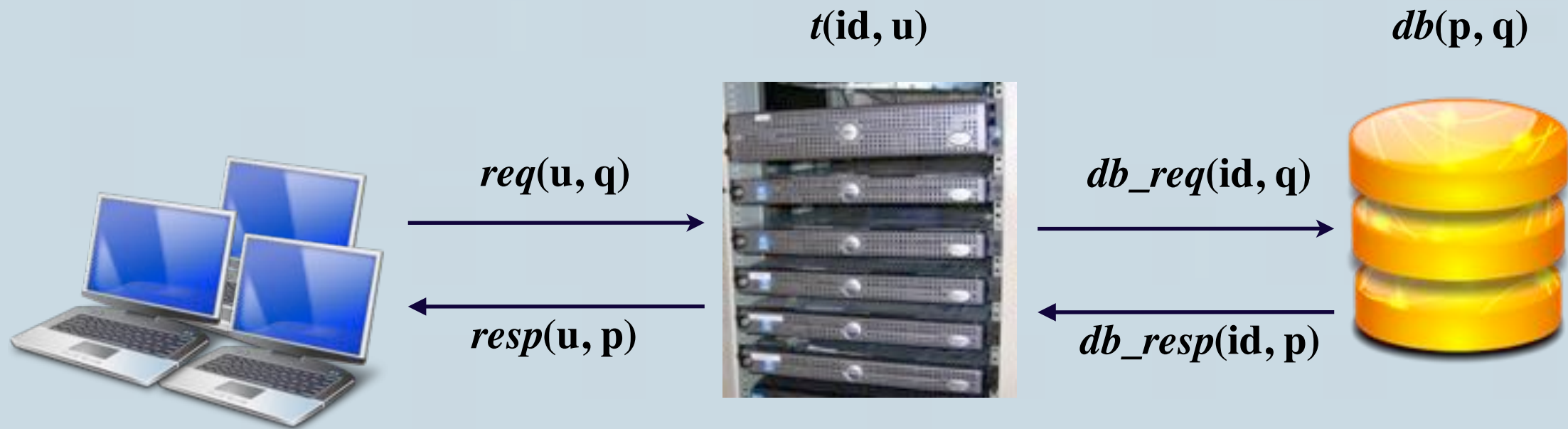


(Simplified)

Invariant: Valid tiling, H closed under successor, f is true.

Encoding: Every valid is finite

Bound I is not complete



Invariant $I = \forall u, p. resp(u, p) \rightarrow \exists q. req(u, q) \wedge db(q, p) \wedge$
 $\forall id, q. db_req(id, q) \rightarrow \exists u. t(id, u) \wedge req(u, q) \wedge$
 $\forall id, p. db_resp(id, p) \rightarrow \exists q. db_req(id, q) \wedge db(q, p) \wedge$
 $\forall id, u1, u2. t(id, u1) \wedge t(id, u2) \rightarrow u1 = u2$

Proof Sketch

$$\mathcal{A} \models \left(\bigwedge_{\vec{c}} \alpha(\vec{c}) \right) \wedge \delta_S \wedge (\neg I')_S$$

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$$\mathcal{A} \models \left(\bigwedge_{\vec{c}} \alpha(\vec{c}) \right) \wedge (\eta(\hat{\delta}))[\psi/r, \psi'/r'] \wedge ((\neg \hat{I}')_S)[\psi/r, \psi'/r']$$

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$$\hat{\mathcal{A}}|_{\hat{C}} \models (\forall \vec{x}. \hat{\alpha}(\vec{x})) \wedge \eta(\hat{\delta}) \wedge (\neg \hat{I}')_S$$

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$$\mathcal{A} \models \left(\bigwedge_{\vec{c}} \alpha(\vec{c}) \right) \wedge \delta_S \wedge (\neg I')_S$$



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$$\hat{\mathcal{A}}|_{\hat{C}} \models \hat{I} \wedge \hat{\delta} \wedge \neg \hat{I}'$$